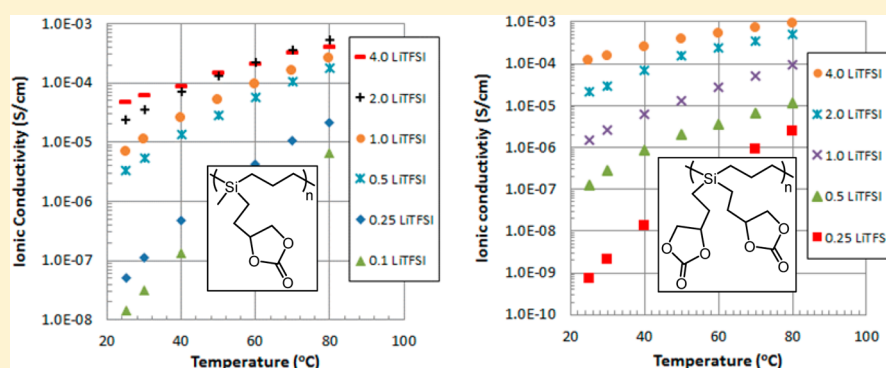


Synthesis and Properties of Polycarbosilanes Having 5-Membered Cyclic Carbonate Groups as Solid Polymer Electrolytes

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S Supporting Information



ABSTRACT: We conducted ring-opening polymerization of silacyclobutane monomers 1-[2-(2,4-dioxo-3-pentanoyl)ethyl]-1-methylsilacyclobutane (SBMC) and 1,1-di[2-(2,4-dioxo-3-pentanoyl)ethyl]silacyclobutane (SBDC) to obtain polySBMC and polySBDC, respectively, and measured their ionic conductivity with addition of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) or lithium trifluoromethanesulfonate (LiOTf). The ionic conductivity of both polySBMC and polySBDC increased with the increase in the amount of LiTFSI added. The ionic conductivity of polySBMC decreased with the increase in the amount of LiOTf, and that of polySBDC decreased after an increase. PolySBMC and polySBDC showed ionic conductivity of 6.1×10^{-5} and 1.5×10^{-4} S/cm at 30 °C, respectively, after the addition of 4 mol equiv of LiTFSI per cyclic carbonate. Differential scanning calorimetry analysis of the polymer revealed that the polymers with a high LiTFSI content formed a crystalline phase around room temperature, but the glass transition temperatures of the amorphous phases were kept low. Infrared analysis suggested the existence of a strong interaction between carbonate carbonyl groups and lithium cations.

■ INTRODUCTION

Solid polymer electrolytes (SPEs) are materials which can transport ions in the solid phase. The utilization of SPEs for lithium ion batteries is considered to be one of the most promising approaches to develop safe and reliable electrical power storage devices.¹ Poly(ethylene oxide)-based solid electrolytes have been widely investigated,² but the ionic conductivities are not sufficient for practical use.

Recently, polymers with carbonate structures have been paid much attention as matrix polymers for SPEs. For instance, Tominaga and co-workers reported that poly(ethylene carbonate)³ and its derivatives,^{4–6} which have a carbonate group in the polymer backbone showed relatively high ionic conductivity in the presence of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI). Mindemark et al. synthesized and examined flexible polycarbonates as SPE.^{7,8} Grinstaff and co-workers also reported on poly(ether 1,2-glycerol carbonate)s.⁹ These findings suggest that soft and flexible polymers having carbonate structures may act as good matrices for SPEs. On the other hand, Sokolov et al. recently

reported that vinyl ether-based copolymers poly(vinyl ethylene carbonate-co-vinyl acetate) bearing 5-membered cyclic carbonate groups in the polymer side chain exhibited higher ionic conductivity in the presence of LiTFSI.¹⁰ Ethylene carbonate, the simplest 5-membered cyclic carbonate, is one of the main components of liquid electrolytes used for usual lithium ion batteries. We anticipate that it may be a good strategy to introduce 5-membered carbonate groups to flexible polymers for creation of novel high performance SPE materials. Synthesis and properties as solid polymer electrolytes of polysiloxanes bearing 5-membered cyclic carbonate groups have already reported by some research groups.^{11–13}

Polycarbosilanes have many unique properties such as high flexibility, excellent thermal stability, chemical and electrochemical stability,^{14,15} so they can be potential key polymers for novel functional materials. We recently reported applications of

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polycarbosilanes to prepare gel polymer electrolytes for lithium ion batteries¹⁶ and to other soft network-forming materials.^{17,18} We have also investigated polycarbosilanes having sugar-derived structures in the polymer side chains and evaluated whether they could be used as biocompatible materials.¹⁹

Polycarbosilanes having 5-membered cyclic carbonate groups in their side chains are expected to be good candidates of new SPE matrices. We synthesized polycarbosilanes having one or two cyclic carbonate groups per repeating unit and examined their ionic conductivity as well as their thermal properties in the presence or absence of lithium salts.

EXPERIMENTAL SECTION

Materials. 1-[2-(2,4-Dioxa-3-pentanoyl)ethyl]-1-methylsilacyclobutane (SBMC) was prepared according to the reported procedure.¹⁶ 1,1-Dichlorosilacyclobutane and platinum(0)-1,3-divinyl-1,1,3,3-tetra-methyldisiloxane complex (solution in xylenes, ~2% platinum) were purchased from Sigma-Aldrich (St. Louis, USA), 4-bromo-1-butene and *m*-Chloroperbenzoic acid (*m*CPBA) from Tokyo Chemical Industry (Tokyo, Japan), 1,2-dibromoethane from Wako Pure Chemical Industry (Osaka, Japan), and magnesium (turnings) from Nacalai Tesque (Kyoto, Japan). These chemicals were used as delivered. Lithium bromide (anhydrous) was purchased from Kanto Chemical Co Inc. (Tokyo, Japan) and dried at 120 °C in vacuo for 2 h before use. CO₂ was purchased from Fukuho Teisan (Iizuka, Japan). Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) was purchased from Wako Pure Chemical Industry, lithium trifluoromethanesulfonate (LiOTf) from Tokyo Chemical Industry, and they were dried in vacuo for 2 h before use. Dichloromethane, ethyl acetate, hexane, acetone, acetonitrile and methanol were purchased from Wako Pure Chemical Industry and used as delivered. Tetrahydrofuran (THF) was purchased from Wako Pure Chemical Industry and freshly distilled over sodium benzophenone ketyl under nitrogen. *N,N*-Dimethylformamide (DMF) was also purchased from Wako Pure Chemical Industry and freshly distilled over calcium hydride before use. CDCl₃, acetone-*d*₆, and dimethyl sulfoxide-*d*₆ (DMSO-*d*₆) for nuclear magnetic resonance measurements were purchased from Cambridge Isotope Laboratories, Inc. (MA, USA). Wakogel C60 was purchased from Wako Pure Chemical Industry and used for silica-gel column chromatography. Lithium ion battery electrolyte solution grade 1 mol/L LiPF₆ in ethylene carbonate (EC)/diethyl carbonate (DEC) (3/7 v/v%) solution was purchased from Kishida Chemical Co., Ltd. (Osaka, Japan). Lithium rod (ϕ , 1 mm; length, 50 mm) for electrode was purchased from Honjo Metal Co., Ltd. (Osaka, Japan).

Characterization. ¹H (400 MHz) and ¹³C (100 MHz) NMR spectra were recorded on a Varian UNITY INOVA400 NMR spectrometer in CDCl₃, DMSO-*d*₆, or acetone-*d*₆. Chemical shifts were determined using tetramethylsilane (TMS) or the residual protons as the internal standards. Infrared spectroscopy (IR) spectra were recorded on a Thermo Scientific Nicolet iS10 spectrometer equipped with a Smart iTR sampling accessory. Number-averaged molecular weight (*M*_n) and weight-averaged molecular weight (*M*_w) were estimated by size exclusion chromatography (SEC) on a TOSOH HLC-8320GPC system equipped with refractive index and ultraviolet (λ = 254 nm) detectors, and three consecutive polystyrene gel columns [TSKgels (bead size, exclusion limited molecular weight); super-AW4000 (6 μ m, >4 $\times$ 10⁵), super-AW3000 (4 μ m, >6 $\times$ 10⁴), and super-AW2500 (4 μ m, >2 $\times$ 10³)] for THF eluent, or polystyrene gel columns [TSKgels super-HM-H (3.0 mm ϕ $\times$ 15 cm, 3 and 5 μ m bead size, >4 $\times$ 10⁸)] for DMF (50 mM LiBr) eluent. The systems were operated at a flow rate of 0.5 mL/min, using THF as an eluent at 40 °C, or operated at a flow rate of 0.6 mL/min, using DMF containing 50 mM LiBr as an eluent at 40 °C. Polystyrene standards were employed for calibration. Differential scanning calorimetry (DSC) was carried out with a Seiko Instrument Inc. DSC-6200 using an aluminum pan under a 20 mL/min N₂ flow at the heating rate of 10 °C/min. Thermal gravimetric analysis (TGA) was performed

with a Seiko Instrument Inc. TG-DTA 6200 using an aluminum pan under a 50 mL/min N₂ flow at a heating rate of 10 °C/min.

Synthesis of 1,1-Di(3-butenyl)silacyclobutane (SBDB). In a 300 mL double-necked round-bottom flask with a three-way stopcock, a reflux condenser, a rubber balloon, a rubber septa, a magnetic stirring bar, and 2.72 g of magnesium (112 mmol) were placed, and the flask was filled with nitrogen. Then 10 mL of THF and 0.3 mL of 1,2-dibromoethane were added to activate the magnesium. A solution of 4-bromo-1-butene (9.5 mL, 94 mmol) in THF (90 mL) was slowly added with stirring, then an exothermic reaction occurred. After the heat generation was over, the mixture was further stirred for 1 h, and then 1,1-dichlorosilacyclobutane (4.7 mL, 40 mmol) was added to the solution, and the whole mixture was stirred for 2 h at 60 °C. The resulting mixture was cooled to room temperature, and poured into ice-cooled water. The organic products were extracted with hexane, and the extracts were washed five times with water, dried over anhydrous Na₂SO₄, and concentrated. Distillation under reduced pressure gave 1,1-di(3-butenyl)silacyclobutane (SBDB) (6.43 g, 35.7 mmol) in 89% yield. Bp: 92–96 °C/5 mmHg. IR (ATR) 752, 907, 991, 1119, 1639 ($\nu_{\text{C}=\text{C}}$), 2913 (ν_{CH_2}) cm⁻¹. ¹H NMR (CDCl₃): δ 0.80–0.92 (m, 4H), 1.01 (t, *J* = 8.2 Hz, 4H), 2.08 (quint, *J* = 8.2 Hz, 2H), 2.14–2.28 (m, 4H), 4.93 (d, *J* = 10.0 Hz, 2H), 5.03 (d, *J* = 16.8 Hz, 2H), 5.92 (ddt, *J* = 16.8, 10.0, 6.4 Hz, 2H) ppm. ¹³C NMR (CDCl₃): δ 12.38 (2-position C of SiC₃ ring), 14.42 (C=C–C–C*–Si), 18.61 (third position C of SiC₃ ring), 27.95 (C=C–C*–C–Si), 113.23 (C*=C–C–C–Si), 141.29 (C=C*–C–C–Si) ppm. Anal. Found: C, 73.33; H, 11.14. Calcd for C₁₁H₂₀Si: C, 73.25; H, 11.18.

Preparation of 1,1-Di(3,4-epoxybutyl)silacyclobutane (SBDEX). SBDB (6.43 g, 35.7 mmol) was charged in a 300 mL round bottomed flask, containing 110 mL of dichloromethane with a magnetic stirring bar. A dichloromethane (25 mL) solution of *m*CPBA (70 wt %, 19.4 g, 78.6 mmol) was added to the mixture at 0 °C. The mixture was stirred at 0 °C for 1 h and then stirred at room temperature for 12 h. The resulting mixture was poured into an aqueous solution of Na₂SO₃. The organic layer was then washed with saturated aqueous NaHCO₃, dried over Na₂SO₄, and concentrated in vacuo. The residual oil was distilled under reduced pressure to give the title compound (5.67 g, 26.7 mmol) in 75% yield. Bp: 98–107 °C (0.1 mmHg). IR (ATR): 725, 739, 758, 831, 903 ($\nu_{\text{C}=\text{O}}$ of epoxy ring), 1119, 2920 (ν_{CH_2}), 2965 (ν_{CH} of epoxy ring) cm⁻¹. ¹H NMR (CDCl₃): δ 0.78–0.94 (m, 4H), 1.01 (t, *J* = 8.2 Hz, 4H), 1.60–1.75 (m, 4H), 2.07 (quint, *J* = 8.2 Hz, 2H), 2.51 (dd, *J* = 2.8, 4.8 Hz, 2H), 2.70 (dd, *J* = 4.0, 4.8 Hz, 2H), 2.92–2.98 (m, 2H) ppm. ¹³C NMR (CDCl₃): δ 10.60 (2-position C of SiC₃ ring), 12.00 (1-position C of 3,4-epoxybutyl), 18.36 (3-position C of SiC₃ ring), 26.71 (2-position C of 3,4-epoxybutyl), 47.18 (3-position C of 3,4-epoxybutyl), 54.06 (4-position C of 3,4-epoxybutyl) ppm. Anal. Found: C, 62.13; H, 9.37. Calcd for C₁₁H₂₀O₂Si: C, 62.21; H, 9.49.

Synthesis of 1,1-Di[2-(2,4-dioxa-3-pentanoyl)ethyl]-silacyclobutane (SBDC). SBDEX (2.80 g, 13.2 mmol) and freshly distilled DMF (20 mL) were charged in a 100 mL round bottomed flask equipped with a three-way stopcock and containing a reflux condenser, a rubber balloon filled with CO₂, a magnetic stirring bar, and lithium bromide (anhydrous) (115 mg, 1.32 mmol). The flask was filled with carbon dioxide, and then kept at 100 °C for 18 h with stirring. Then the mixture was cooled to room temperature and poured into water. The products were extracted with ethyl acetate (200 mL), dried over Na₂SO₄, and concentrated in vacuo. Purification of the residual oil by silica-gel column chromatography gave 1,1-di[2-(2,4-dioxa-3-pentanoyl)ethyl]silacyclobutane (SBDC) (3.09 g, 10.3 mmol) as a colorless solid in 78% yield. Mp: 36.0–39.0 °C. IR (ATR) 718, 773, 1061 ($\nu_{\text{C}=\text{O}}$ of carbonate), 1161 ($\nu_{\text{C}=\text{O}}$ of carbonate), 1790 ($\nu_{\text{C}=\text{O}}$), 2922 (ν_{CH_2}) cm⁻¹. ¹H NMR (CDCl₃): δ (ppm) 0.76–0.88 (m, 2H), 0.88–0.98 (m, 2H), 1.07 (t, *J* = 8.4 Hz, 4H), 1.80–1.96 (m, 4H), 2.11 (quint, *J* = 8.4 Hz, 2H), 4.15 (dd, *J* = 7.0, 8.4 Hz, 2H), 4.58 (dd, *J* = 8.2, 8.4 Hz, 2H), 4.75 (dddd, *J* = 6.7, 6.7, 7.0, 8.2 Hz, 2H) ppm. ¹³C NMR (CDCl₃): δ 9.08 (2-position C of SiC₃ ring), 11.74 (1-position C of ethyl), 18.42 (3-position C of SiC₃ ring), 28.13 (2-position C of ethyl), 68.97 (5-position C of 2,4-dioxa-3-pentanoyl),

78.48 (1-position C of 2,4-dioxo-3-pentanoyl), 155.06 (C=O) ppm. Anal. Found: C, 52.04; H, 6.59. Calcd for $C_{13}H_{20}O_6Si$: C, 51.98; H, 6.71.

Polymerization of SBDC. SBDC (600 mg, 2.0 mmol) was placed in a 100 mL round bottomed flask equipped with a magnetic stirring bar, three-way stopcock, and a rubber balloon filled with nitrogen, and the flask was filled with nitrogen. Then ethyl acetate (2.0 mL) and 40 mg of platinum(0)-1,3-divinyl-1,1,3,3-tetramethyldisiloxane complex (solution in xylenes, $\sim 2\%$ platinum, 4.0 μ mol) were added, and the mixture was heated at 80 $^{\circ}$ C. After several minutes, polymeric products were precipitated. The mixture was heated at 80 $^{\circ}$ C for 2 h with stirring to complete the polymerization. The resulting mixture was cooled to room temperature. Precipitated compounds were collected and dried at 70 $^{\circ}$ C in vacuo to give polySBDC (590 mg) in 98% yield. 1H NMR (DMSO- d_6): δ 0.38–0.78 (m, 8H), 1.20–1.37 (m, 2H), 1.64 (q, J = 7.7 Hz, 4H), 4.12 (dd, J = 7.2, 7.2 Hz, 2H), 4.55 (dd, J = 7.2, 8.4 Hz, 2H), 4.71 (dddd, J = 7.0, 7.2, 7.7, 8.4 Hz, 2H) ppm. M_n = 90 600 g/mol, M_w = 270 000 g/mol, M_w/M_n = 2.51.

Measurement. A linear sweep voltammogram (LSV) was taken in a tripolar electrode beaker cell (counter electrode: platinum, working electrode: glassy carbon of 3 mm diameter, reference electrode: lithium) with a Hokuto Denko HABF-501A potentiostat/galvanostat at a sweep rate of 10 mV/s in a potential range from the oxidation side: OCP $\sim$ 7.5 V to the reduction side: OCP $\sim$ -1 V. The ionic conductivity was measured by HIOKI 3532-80 chemical impedance meter at 50 mV (frequency range of 4 Hz to 100 kHz) using stainless coin cell with Teflon guide (ϕ = 16 mm, Eager Corp., Japan) according to the complex impedance analysis. The samples for the ionic conductivity measurements were prepared by casting an acetone solution (for polySBMC) and an acetonitrile solution (for polySBDC) of polymer with lithium salt followed by evaporating the solvent and vacuum-dried for 48 h.

RESULTS AND DISCUSSION

Synthesis of Polycarbosilanes Having One or Two 5-Membered Cyclic Carbonate Groups per Repeating Unit. We prepared two types of polycarbosilanes bearing 5-membered cyclic carbonate groups shown in Figure 1. One was

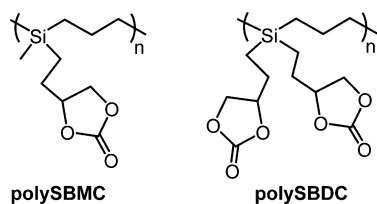
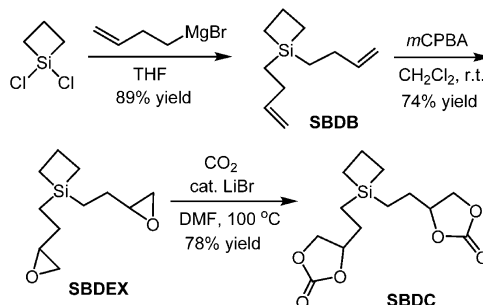


Figure 1. Chemical structures of polycarbosilanes having 5-membered cyclic carbonate groups.

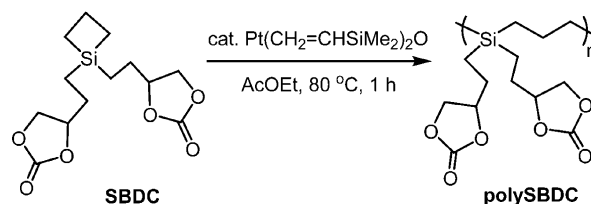
polySBMC, which had one 5-membered cyclic carbonate group per repeating unit. This polymer was synthesized by Pt-complex-catalyzed ring-opening polymerization of a silacyclobutane derivative having one cyclic carbonate group (SBMC). SBMC monomer and its polymer were prepared as reported previously.¹⁶ Number-averaged molecular weight (M_n) and weight-averaged molecular weight (M_w) relative to polystyrene were 17 300 g/mol, and 22 600 g/mol, respectively. The other polymer was polySBDC, which had two 5-membered cyclic carbonate groups per repeating unit. PolySBDC was synthesized from SBDC. The SBDC monomer was prepared in three steps from a commercially available 1,1-dichlorosilacyclobutane according to the synthetic route shown in Scheme 1. 1H NMR spectrum of SBDC is given in Figure S1(a). Six protons on the cyclic carbonate ring a, b, c were observed at 4.75, 4.58, and 4.15 ppm. Six protons on the 4-membered sila-carbocycle were observed at 2.11 and 1.07 ppm, respectively. Polymerization of

Scheme 1. Synthesis of SBDC



SBDC was conducted with a platinum complex catalyst as shown in Scheme 2, using ethyl acetate as a solvent, because

Scheme 2. Polymerization of SBDC



SBDC monomer was too viscous to be polymerized in bulk. 1H NMR spectrum of polySBDC is depicted in Figure S1(b). Signals of methylene protons on the 3-position of silacyclobutane, which were observed at 2.11 ppm in Figure S1(a) appeared at around 1.25 ppm as the open-ring methylene protons in Figure S1(b), while the signals for 5-membered cyclic carbonate groups remained intact. These observations clearly indicated that the ring-opening polymerization of silacyclobutane ring selectively proceeded with the Pt-complex catalyst keeping the 5-membered cyclic carbonate ring unchanged. M_n and M_w relative to polystyrene of the obtained polySBDC were 90 600 g/mol and 227 000 g/mol, respectively, which were higher than those of polySBMC. Cyclic carbonate groups may coordinate to Pt catalyst site on the propagating chain end to suppress β -H elimination. Since the local concentration of the carbonate moiety around the active in polySBDC is higher than in polySBMC, the β -H elimination in the polymerization of SBDC is more suppressed than that of SBMC. We consider that this may be the reason for the higher molecular weight of polySBDC. GPC charts for polySBMC and polySBDC synthesized here are given in Figure S2.

Electrochemical Stability. To apply these polymers to solid polymer electrolytes, it is important that they are electrochemically stable. We previously reported the high electrochemical stability (0–4.2 V) of polySBMC,¹⁶ which was measured by linear sweep voltammetry. To examine the electrochemical stability of newly synthesized polySBDC, we examined LSV of polySBDC. Figure 2 shows the LSV curve for polySBDC in 1 mol/L $LiPF_6$ /ethylene carbonate (EC)/diethyl carbonate (DEC) (3/7). No oxidation peak was observed below 4.6 V, and no reduction peak was observed above -0.3 V, indicating that polySBDC has also high electrochemical stability of and that the polymer can be used in usual lithium ion batteries.

Ionic Conductivity Measurement. Figure 3 shows the ionic conductivity as a function of temperature for the polySBMC after the addition of (a) LiOTf and (b) LiTFSI.

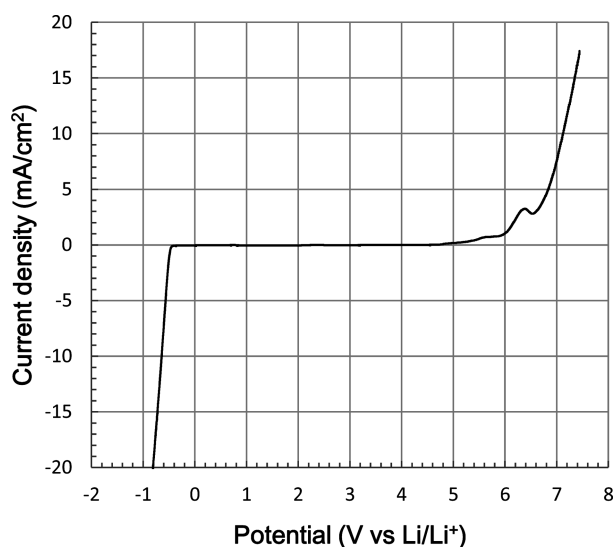


Figure 2. Cyclic voltammogram of 1 wt % polySBDC in 1 mol/L $\text{LiPF}_6\text{-EC/DEC}$ (3/7).

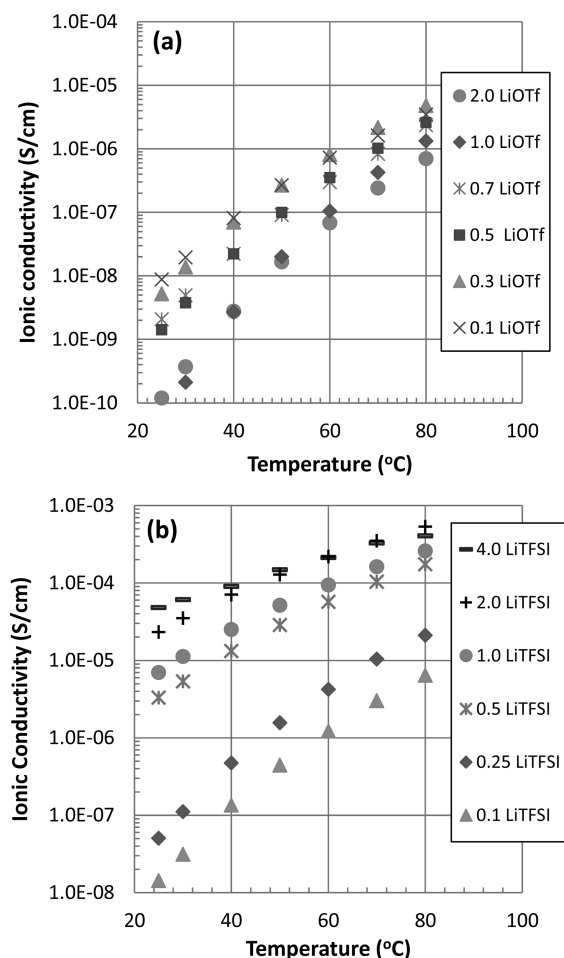


Figure 3. Ionic conductivity as a function of temperature for polySBMC with the addition of (a) LiOTf and (b) LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

The amount of lithium salt added was denoted by a mole ratio of the lithium salt to the carbonate group. The ionic conductivity in the presence of LiOTf was low in general,

and the conductivity decreased with the increase of the amount of LiOTf added. The maximum ionic conductivity measured at 30 °C was 1.95×10^{-8} S/cm when 0.1 mol equiv of LiOTf was added to the carbonate group. On the other hand, the ionic conductivity was higher in the presence of LiTFSI than in that of LiOTf, and it increased monotonically with the increase of added LiTFSI. The same trend has been previously reported separately by Tominaga³ and Sokolov.¹⁰ The maximum ionic conductivity measured at 30 °C for polySBMC was 6.11×10^{-5} S/cm when 4.0 equiv of LiTFSI was added to the carbonate group. Further addition of LiTFSI to the polymer made it difficult to prepare homogeneous samples suitable for ionic conductivity measurements. We assume that the conductivity differences between LiTFSI and LiOTf may be due to their solubility difference toward the polymer, though the details of their solubility to polymers having carbonate groups have not clarified yet. The high solubility of LiTFSI to polySBMC may make it possible to form the “polymer in salt electrolyte” system, proposed by Angell and co-workers.²⁰ Figure 4 shows the ionic conductivity for the polySBDC after addition of (a) LiOTf or (b) LiTFSI. The ionic conductivity of polySBDC in the presence of LiOTf once increased but decreased later with increasing the amount of LiOTf. On the other hand, the ionic conductivity increased monotonically with the increase of

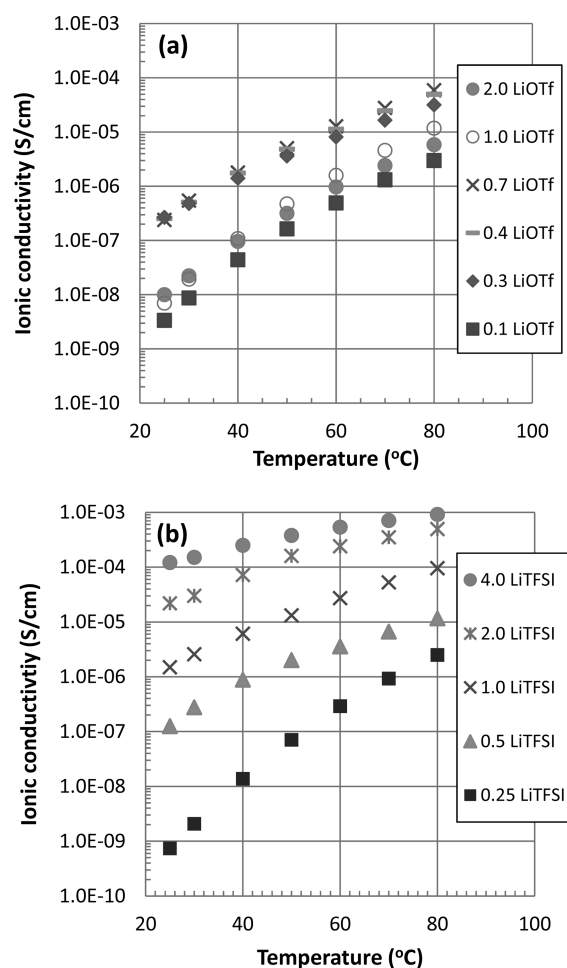


Figure 4. Ionic conductivity as a function of temperature for polySBDC with the addition of (a) LiOTf and (b) LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

LiTFSI. The ionic conductivity of polySBDC with LiTFSI was higher than that of polySBMC with LiTFSI. It is worth noting that the ionic conductivity of polySBDC in the presence of 4.0 mol equiv of LiTFSI per carbonate group reached 1.21×10^{-4} S/cm at 30 °C, which was significantly high compared from the reported values,^{3,10} for solid polymer electrolytes. Typical photographs of polySBMC and polySBDC added with 4.0 equiv of LiTFSI per cyclic carbonate group are given in Figure S3.

DSC Analysis. Figure 5a shows the DSC curves of the heating scan for polySBMC with or without addition of LiOTf. Neat polySBMC exhibited a relatively low glass-transition temperature (T_g) of −20 °C, but did not show any melting transition in the measured temperature range. The T_g of polySBMC hardly changed and remained at around −20 to −25 °C in the presence of LiOTf. It should be noted that a

large endothermic transition peak appeared at 30–50 °C for polySBMC in the presence of 0.2 mol equiv of LiOTf, and a smaller endothermic transition was also observed from 50 to 56 °C for polySBMC in the presence of 0.5 mol equiv of LiOTf, which were probably due to the melt transition of LiOTf-complexed polySBMC segment. This peak was not observed in the curve of polySBMC in the presence of 1.0 mol equiv of LiOTf. Figure 5b shows the DSC curves for polySBMC with or without LiTFSI. The T_g of the polymer hardly changed and was observed at around −20 °C regardless of the amount of LiTFSI added. However, polySBMC in the presence of 4.0 mol equiv of LiTFSI showed a large endothermic peak at 30–45 °C, which may be a melting transition of LiTFSI-complexed polySBMC segment.

Figure 6a shows the DSC curve for polySBDC with or without LiOTf. Neat polySBDC also exhibited a relatively low

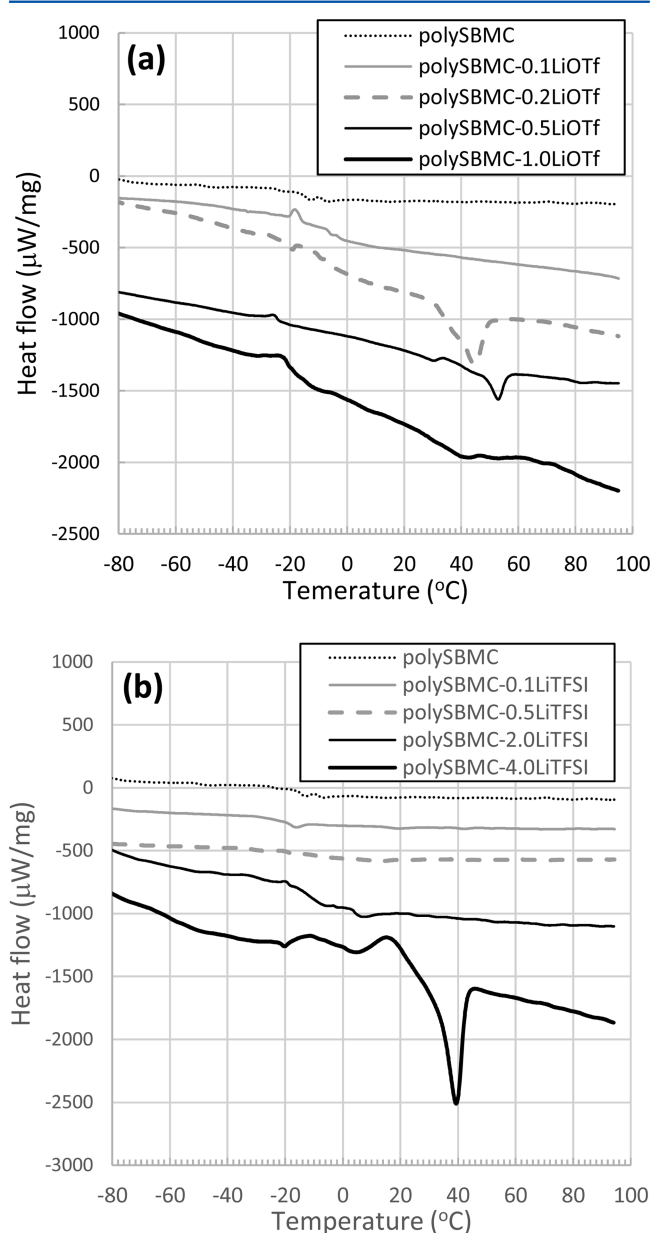


Figure 5. DSC heating scan curves for polySBMC with or without (a) LiOTf and (b) LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

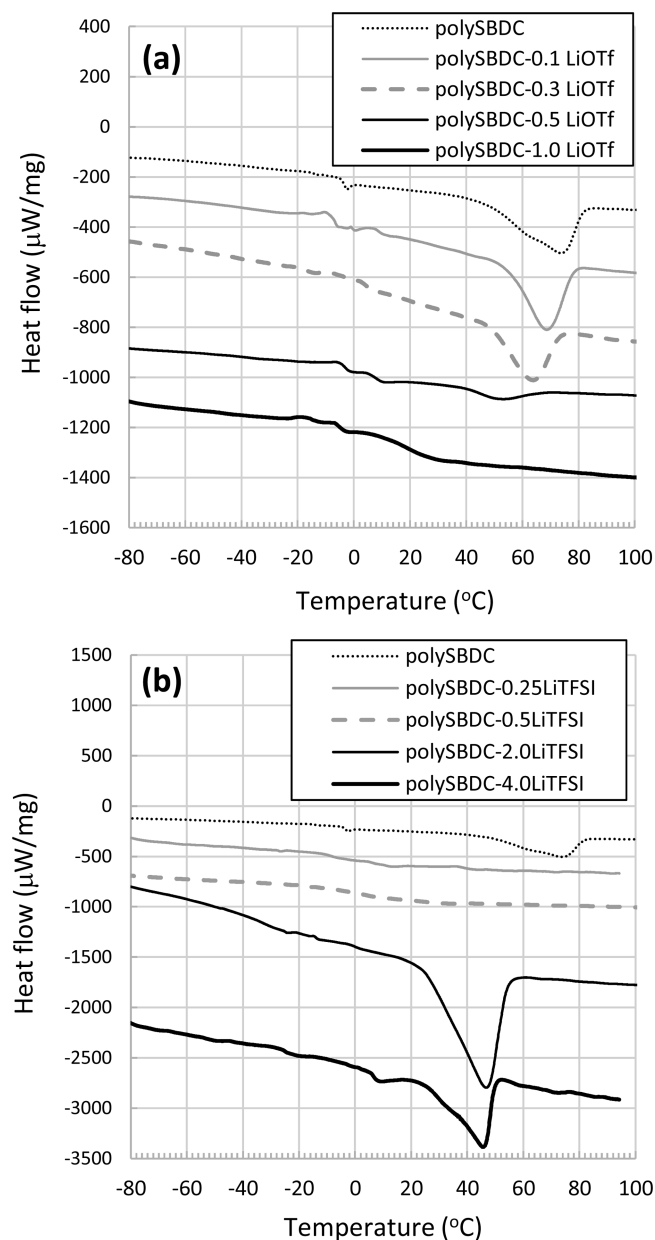


Figure 6. DSC heating scan curves for polySBDC with or without (a) LiOTf and (b) LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

glass-transition temperature (T_g) of $-5\text{ }^{\circ}\text{C}$, though it had two bulky 5-membered cyclic carbonate groups per repeating unit. In contrast to polySBMC, it showed a melting transition in the range of $40\text{--}85\text{ }^{\circ}\text{C}$, which indicated that this polymer is a semicrystalline polymer. After the addition of a small amount of LiOTf (<0.3 mol equiv), the melting transition peak remained but was slightly shifting to a lower temperature region ($40\text{--}72\text{ }^{\circ}\text{C}$). The T_g of polySBDC hardly changed and remained at around -5 to $-15\text{ }^{\circ}\text{C}$ after the addition of LiOTf. We supposed that the endothermic peak may be due to the melting of LiOTf-complexed polySBDC segments. Figure 6b depicted DSC curves for polySBDC with or without addition of LiTFSI. The melting transition disappeared in the presence of a small amount of LiTFSI (<0.5 mol equiv), but a large endothermic peak appeared in the presence of LiTFSI (>2.0 mol equiv of LiTFSI). This large peak may be attributed to the melting transition of LiTFSI-complexed polySBDC segments. We have not yet fully understood the thermal phase transition at this stage, and further experiments must be carried out. It should be noted that the T_g of polySBDC decreased with the increase in the amount of LiTFSI added (from $-5\text{ }^{\circ}\text{C}$ for neat polySBC to $-25\text{ }^{\circ}\text{C}$ for polySBDC with 4.0 mol equiv of LiTFSI). These results suggest that both polySBMC and polySBDC form a kind of crystalline phase after addition of a large amount of LiTFSI while keeping the T_g amorphous polymer phase low. However, we consider that the ionic conductivity was hardly affected by the crystalline phase. When a large amount of LiTFSI was added to the polymers, the ionic conductivity of the polymer showed quite high ionic conductivity, for example $1.2 \times 10^{-4}\text{ S/cm}$ at $30\text{ }^{\circ}\text{C}$ of polySBDC with 4.0 LiTFSI. We consider that one reason for this high ionic conductivity may be these T_g s of the polymers, but we also assume that in these cases ion conduction may be decoupled with the segmental motion of the polymers. In fact, temperature dependence of the ionic conductivity in the case of high salt content is much weaker than that of low salt content sample as shown in Figures 2b and 3b. This may also be the reason why ionic conductivity of polySBDC is higher than that of polySBMC at the same LiTFSI content despite the higher T_g value. The relationship between the polymer phases and ionic conductivity as well as the ion conduction mechanisms at this stage remains to be elucidated further.

IR Spectroscopy Analysis. Parts a and b of Figure 7 show expanded IR spectra ($1400\text{--}2000\text{ cm}^{-1}$) of polySBMC taken with or without LiOTf or LiTFSI added. The absorption of the $\text{C}=\text{O}$ stretching of the carbonate group was observed at 1790 cm^{-1} for polySBMC without the addition of lithium salt, and it shifted to 1770 cm^{-1} after addition of LiOTf or LiTFSI. Similar absorption shifts of the carbonyl $\text{C}=\text{O}$ stretching were observed also in both LiOTf-added and LiTFSI-added polySBDC. The IR spectra for polySBDC with lithium salts are given in Figure S4. These observations clearly indicate a strong interaction between the carbonate carbonyl in those polymers and the added lithium cations. The same phenomena of IR absorption shifts were previously reported by others.^{5,8}

Furthermore, expanded IR spectra ($700\text{--}900\text{ cm}^{-1}$) of LiTFSI, polySBMC, and polySBMC with LiTFSI are depicted in Figure 8. A $\text{C}\text{--}\text{F}$ bending absorption of CF_3 group (δ_{CF_3}) was observed at 742 cm^{-1} in polySBMC with 1.0 LiTFSI, while that corresponds to neat LiTFSI was observed at 747 cm^{-1} . The same IR shift was observed in the case of polySBDC (Figure S5). These IR shifts clearly indicated that the LiTFSI

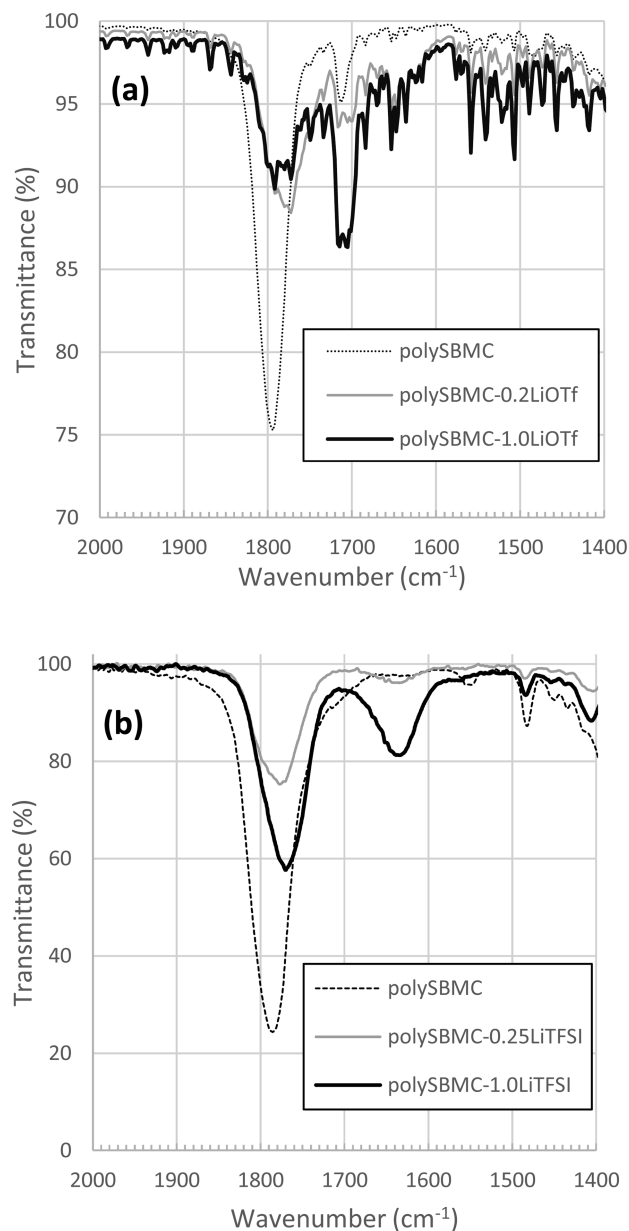


Figure 7. Expanded IR spectra ($1400\text{--}2000\text{ cm}^{-1}$) of polySBMC with or without (a) LiOTf and (b) LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

salt highly dissociated by addition of polycarbosilanes having carbonate groups. The same phenomenon was previously reported in the Raman spectroscopy.²¹

CONCLUSIONS

In this study, we synthesized polycarbosilanes having one or two 5-membered cyclic carbonate groups per repeating unit, and examined their characteristics as solid polymer electrolytes by addition of lithium salts. We found that the ionic conductivity of these polymers increased with the increase in the amount of added LiTFSI. The ionic conductivity at $30\text{ }^{\circ}\text{C}$ was 6.1×10^{-5} and $1.5 \times 10^{-4}\text{ S/cm}$, respectively for the polycarbosilanes polySBMC and polySBDC in the presence of 4 mol equiv of LiTFSI per carbonate group. These polymer/salt systems may form a kind of “polymer in salt electrolytes” to exhibit rather high ionic conductivity. DSC measurements

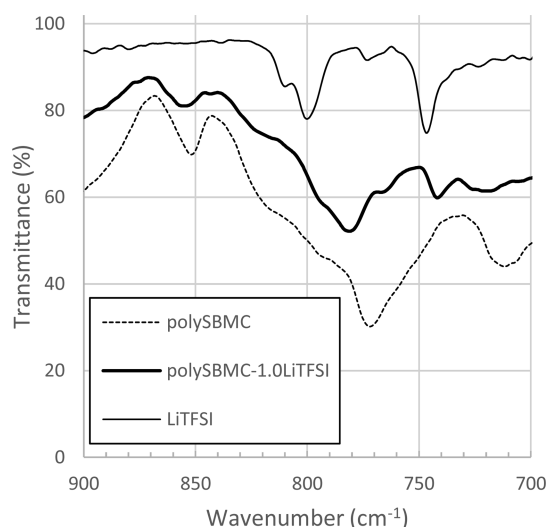


Figure 8. Expanded IR spectra (700–900 cm^{-1}) of LiTFSI, polySBMC, and polySBMC with 1.0 LiTFSI. Amounts of the lithium salt added are denoted by mole ratios of lithium salt to carbonate group.

revealed that in the presence of a large amount of LiTFSI, both polySBMC and polySBDC formed crystalline phases with a melting temperature ranging from 30 to 75 $^{\circ}\text{C}$, while keeping the T_g of amorphous phase low. The thermal stabilities of the polymers was diminished by the addition of lithium salt, but the stability was suitable for practical applications. Although the particular ion-transporting mechanism for this electrolyte system remains unknown, the polymers with 5-membered cyclic carbonate groups can be promising candidates as high performance solid polymer electrolytes. Further investigations on the properties of these polymers are currently in progress.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.6b01516.

^1H NMR spectra for SBDC and polySBDC taken in DMSO- d_6 (Figure S1); GPC charts of polySDMC and polySBDC (Figure S2); photographs of polySBMC and polySBDC added with 4.0 equiv of LiTFSI per cyclic carbonate group (Figure S3); expanded IR spectra (Figures S4 and S5) (PDF)

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Notes

The authors declare no competing financial interest.

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